What AI Sees That We Miss: Aneurysm Detection in Stroke Imaging

Background and Purpose: In acute cerebral infarction, clinical attention is naturally directed toward the causative stenosis or occlusion—a form of confirmation bias that may cause co-existing pathologies such as intracranial aneurysms to be overlooked. We present a case in which an AI-based cerebrovascular analysis platform (CLARUS-N) identified an incidental aneurysm during stroke workup.

Case Description: A 70-year-old male presented with acute dysarthria and motor dysphasia. Brain MRI and MRA confirmed a left MCA territory infarction with near-total left M1 occlusion. The patient was admitted to the ICU and started on dual antiplatelet therapy. Standard radiology interpretation focused on the culprit M1 lesion. Upon application of the CLARUS-N AI pipeline—which simultaneously analyzes both stenosis and aneurysm on MRA—an additional aneurysmal finding was identified that had not been highlighted in the primary report.

Discussion: Clinicians evaluating acute stroke imaging operate under significant cognitive and time pressure, anchoring to the dominant lesion while anatomically distinct findings may go undetected. Unlike human readers, AI systems do not anchor to a primary diagnosis; they evaluate the entire vascular tree with uniform attention across pathology categories. AI-assisted analysis thus functions as a cognitive complement—compensating for attentional bias and enabling comprehensive vascular assessment even when clinical context predisposes the reader toward a narrow diagnostic focus.

Conclusion: In patients presenting with cerebral infarction, confirmation bias predictably narrows clinician attention to stenotic lesions, leaving co-existing aneurysms at risk of being missed. AI-based cerebrovascular analysis, by simultaneously and impartially evaluating both pathologies, offers a practical safeguard. Broader integration of such tools into acute stroke workflows may meaningfully improve diagnostic completeness and patient safety.

Keywords: cerebral infarction, intracranial aneurysm, confirmation bias, AI-assisted diagnosis, MRA, CLARUS-N